DA627 Final presentation

-Saptarshi Mukherjee(220150019)

Contents

- Pipeline(recap)
- Datasets used for experiments
- Experimental results:

  (Recall(Hit@1, 5, 10), LLM(Phi-3.5 vs Llava-1.5), Encoder(ViT-B/16 vs ViT-B/32), RAG vs no RAG)

- Sample demo using captions from both the datasets
- Future studies

Pipeline(recap)

- Load in CLIP encoding model (Example: ViT-B-32)
- Load captions into memory for faster responses
- Compute caption embeddings using the CLIP model
- Given user image, compute image embeddings
- Retrieve top k captions using cosine similarity between image and text embeddings
- Pass in the image and the retrieved captions to the LLM to get the RAG -enabled response(For no RAG, omit the caption retrieval steps)

LLM Prompt

Prompt format:

```
<|user|>
<|image_1|>
Describe this image in one sentence.

Helpful retrieved captions:
- caption_1
- caption_2
...
- caption_k

Only answer with the description sentence.
<|end|>
<|assistant|>
```

The prompt and image are passed to the processor to obtain tokenized inputs and image embeddings. These are then fed into the vision-language model for generation.

Generation steps:

- The model generates output tokens autoregressively.
- Greedy decoding is used to obtain deterministic outputs.
- The maximum number of generated tokens is limited to 25 for better alignment with the ground truth captions(saner experiment results)

Datasets used for experiments:

Flickr-30:

Scale and structure: ~31K images with ~158K captions (≈5 captions per image), providing multiple natural language descriptions per scene—useful for building a diverse caption embedding space.

Coreference and entity annotations: ~244K coreference chains link equivalent mentions across captions, enabling grouping of semantically similar phrases during retrieval.

Ground-truth captions for the same image:

- A fashionably dressed woman holding a record in a frame at a used bookstore .
- A woman holds a framed piece of art at a store while another man shops .
- A girl and a guy shop for records in a music store .
- A man and woman look at music cds .
- Shoppers in a record store .

MSCOCO dataset:

Captions for this image:

1. A black Honda motorcycle parked in front of a garage.
2. A Honda motorcycle parked in a grass driveway
3. A black Honda motorcycle with a dark burgundy seat.
4. Ma motorcycle parked on the gravel in front of a garage
5. A motorcycle with its brake extended standing outside

Dataset usage note: Used the val2017 subset of MS COCO (~5K images) instead of the full dataset to enable faster training and stay within Kaggle GPU and memory constraints.

Compact but diverse caption set: ~5K images with ~5 captions each, offering multiple natural language descriptions per image—useful for building a varied and robust caption embedding pool for retrieval.

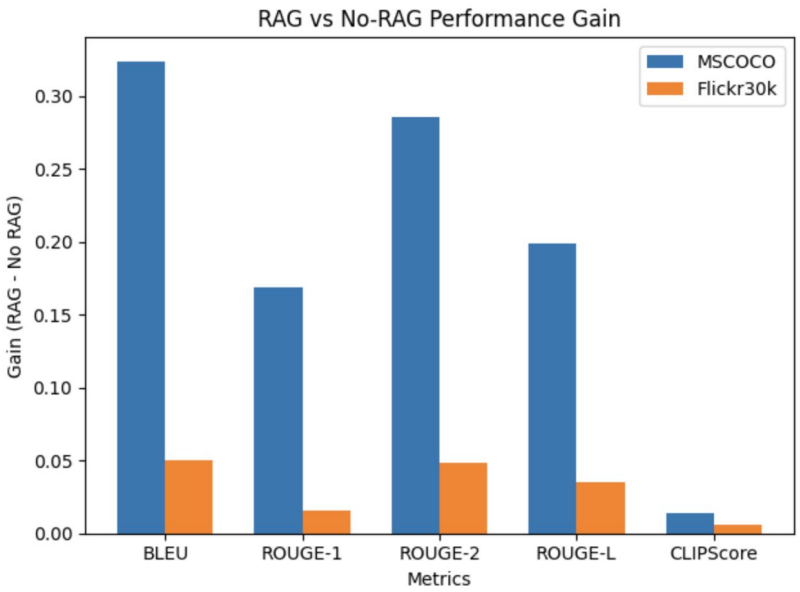
High-quality, real-world descriptions: Human-written captions capture complex everyday scenes with diverse phrasing, helping improve semantic matching and generalization in the multimodal RAG pipeline.

Metrics

- **BLEU:** n-gram precision (0–1) → higher = more exact wording overlap; strong gains show better caption accuracy.
- **ROUGE-1:** unigram recall (0–1) → higher = better coverage of key objects/concepts.
- **ROUGE-2:** bigram recall (0–1) → higher = better phrase-level coherence.
- **ROUGE-L:** sequence overlap (0–1) → higher = more fluent, well-structured sentences.
- **CLIPScore:** image–text semantic similarity (~0–1 typical) → higher = better alignment with visual content.

Experimental results:

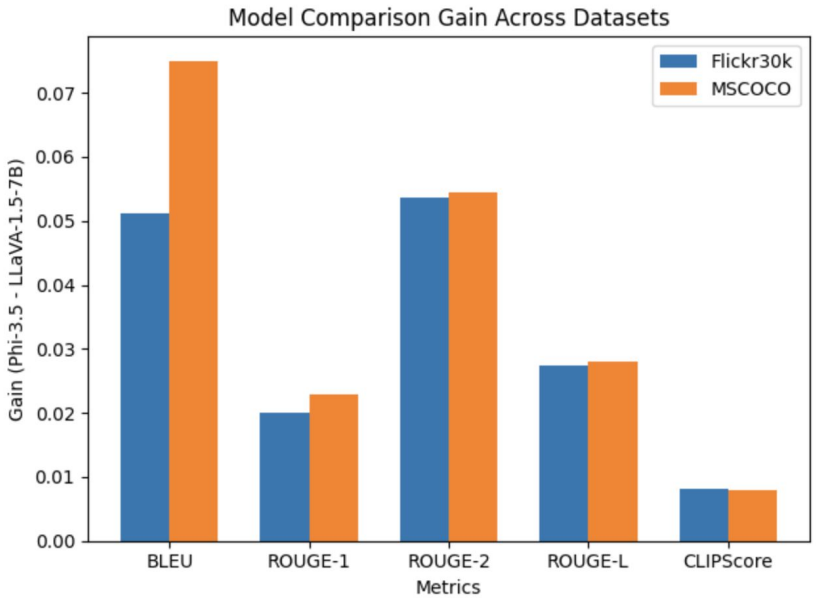
RAG vs no RAG:



	Flickr-30k		MSCOCO	
Metrics	RAG	NO RAG	RAG	NO RAG
BLEU	0.256	0.206	0.539	0.215
ROUGE-1	0.562	0.546	0.751	0.582
ROUGE-2	0.351	0.303	0.612	0.327
ROUGE-L	0.531	0.496	0.727	0.528
CLIPScore	0.327	0.321	0.339	0.324

RAG consistently improves performance across all metrics, with significantly larger gains on MS COCO, highlighting the value of richer and more diverse caption retrieval.

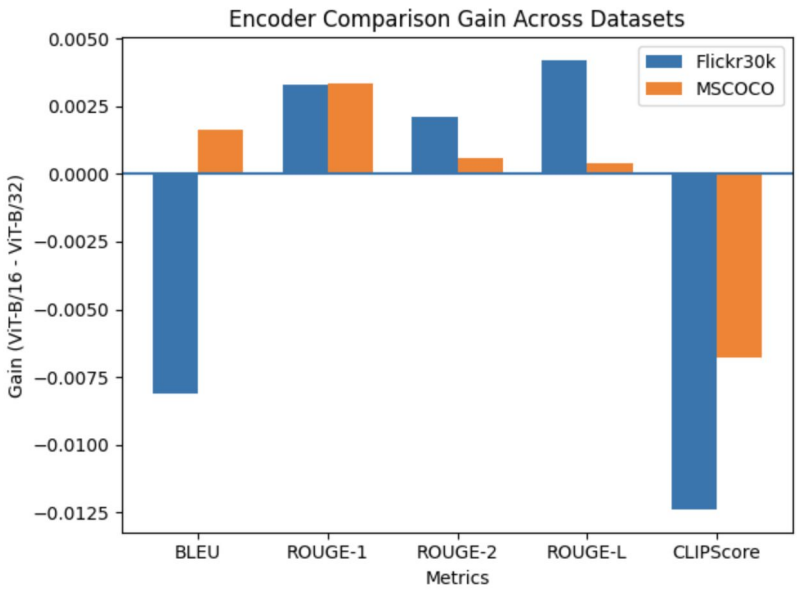
LLM study: Phi-3.5-vision-instruct vs Llava-1.5b-7b-hf



	Flickr-30k		MSCOCO	
Metrics	Phi-3.5	Llava-1.5	Phi-3.5	Llava-1.5
BLEU	0.423	0.372	0.509	0.434
ROUGE-1	0.679	0.659	0.730	0.707
ROUGE-2	0.521	0.468	0.577	0.523
ROUGE-L	0.655	0.628	0.703	0.675
CLIPScore	0.344	0.336	0.339	0.331

There is a statistically significant difference(at P=0.05) between the the answer quality of the 2 models, hence we can say that Phi-3.5 is better, and the gain is similar for both datasets.

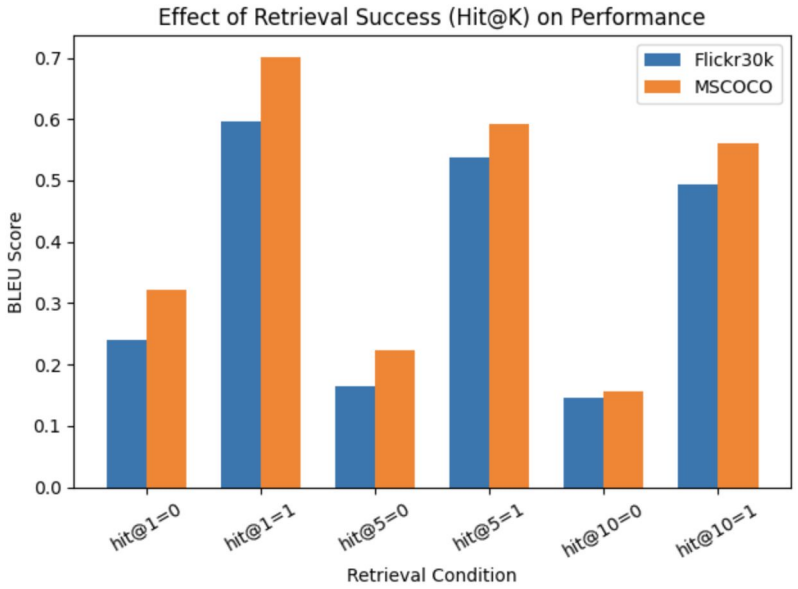
Encoder study: ViT-B/16 vs ViT-B/32



	Flickr-30k		MSCOCO	
Metrics	ViT-B/16	ViT-B/32	ViT-B/16	ViT-B/32
BLEU	0.2473	0.2554	0.5097	0.5080
ROUGE-1	0.5651	0.5618	0.7314	0.7281
ROUGE-2	0.3529	0.3508	0.5707	0.5702
ROUGE-L	0.5351	0.5308	0.7025	0.7021
CLIPScore	0.3143	0.3267	0.3291	0.3359

The difference between performance using ViT-B/16, ViT-B/32 is not statistically significant-very similar performance for both datasets.

Recall study:



	Flickr-30k(200 samples)		MSCOCO(200 samples)	
BLEU	Count	Score	Count	Score
Hit@1=0	101	0.241	95	0.322
Hit@1=1	99	0.596	105	0.701
Hit@5=0	65	0.164	39	0.223
Hit@5=1	135	0.538	161	0.593
Hit@10=0	44	0.146	20	0.156
Hit@10=1	156	0.493	180	0.561

- We observe that if the retrieved captions do not contain atleast one ground truth caption, the answer is of worse quality.
- Retrieving more captions actually worsens quality-makes the model more confused about the wording

Sample demo using both the datasets

```
Top retrieved captions:

0.321 | A black Honda motorcycle with a dark burgundy seat.
0.300 | A very nice looking motorcycle with some long handle bars.
0.296 |  A black motorcycle parked in a parking lot in front of a grass filled hill .
0.291 | A black Honda motorcycle parked in front of a garage.
0.287 | Large black motorcycle sitting next to a white building.

Ground-truth captions for the same image:

- A black Honda motorcycle parked in front of a garage.
- A Honda motorcycle parked in a grass driveway
- A black Honda motorcycle with a dark burgundy seat.
- Ma motorcycle parked on the gravel in front of a garage
- A motorcycle with its brake extended standing outside
```

```
Answer:
 A black Honda motorcycle with a dark burgundy seat is parked in front of a garage.
```

Future studies

- Improve retrieval (reranking to boost Hit@K)
- Adaptive top-k instead of fixed k
- Fine-grained (entity/phrase-level) retrieval
- Test stronger / different retrievers
- Scale to larger datasets (full MS COCO)
- Add human evaluation

Link to the github repository

https://github.com/zombie-programmer-code/DA627_course_project